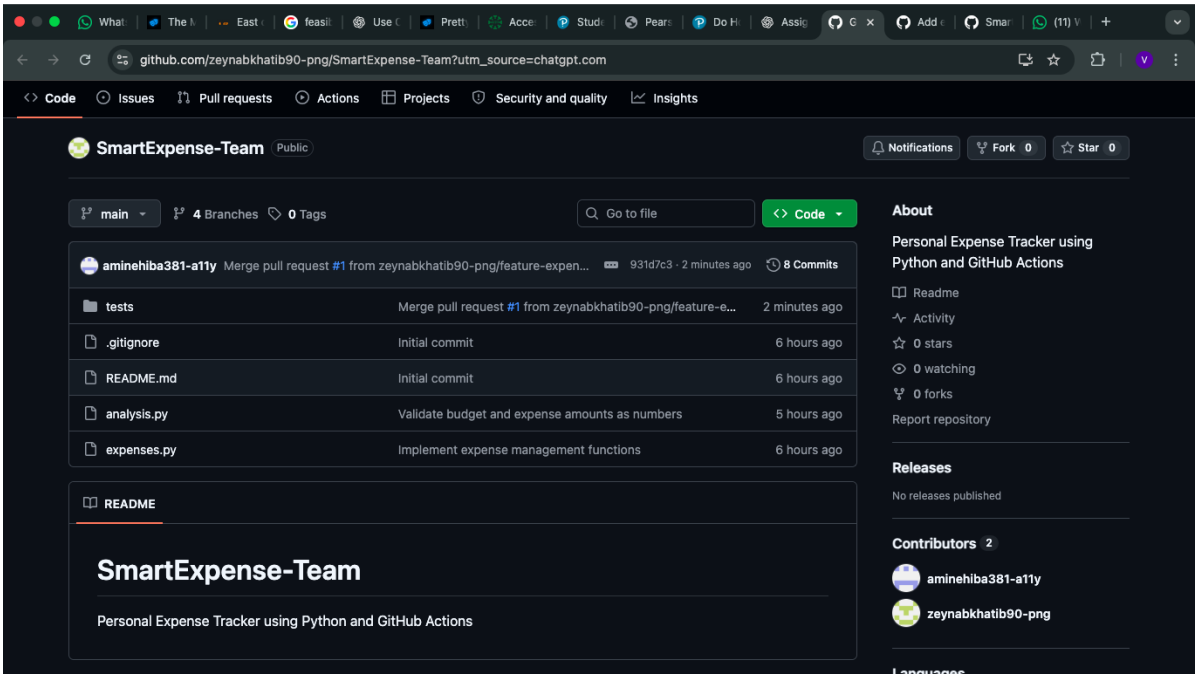
SmartExpense

Assignment 2 - GitHub Collaboration & Automation
Session 2 - Group Lab Work | 26 October 2026

Group size	2 students
Theme	Expense Tracker
Member 1	Zeynab Khatib | ID 146586 | GitHub: zeynabkhatib90-png
Member 2	Hiba Ait Salah | ID 147272 | GitHub: aminehiba381-a11y
Course code / section	CSC332402
Repository	https://github.com/zeynabkhatib90-png/SmartExpense-Team
Published site	https://zeynabkhatib90-png.github.io/SmartExpense-Team/

1. Project & Role Distribution

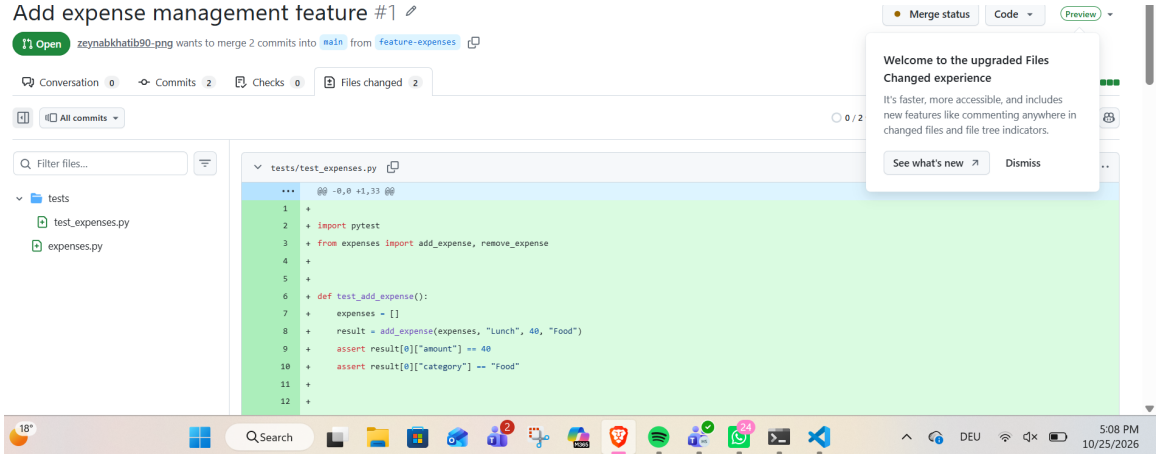
The group followed the two-member pathway. SmartExpense is a Python expense tracker with two complementary components. Zeynab developed expense addition, deletion and input validation in **expenses.py**. Hiba developed totals, category spending, highest-expense reporting and budget analysis in **analysis.py**. A static GitHub Pages site describes the project; no database or server is required.



Evidence 1 - Shared repository on main with Python modules, tests and two contributors.

2. Branches, Commits & Contributions

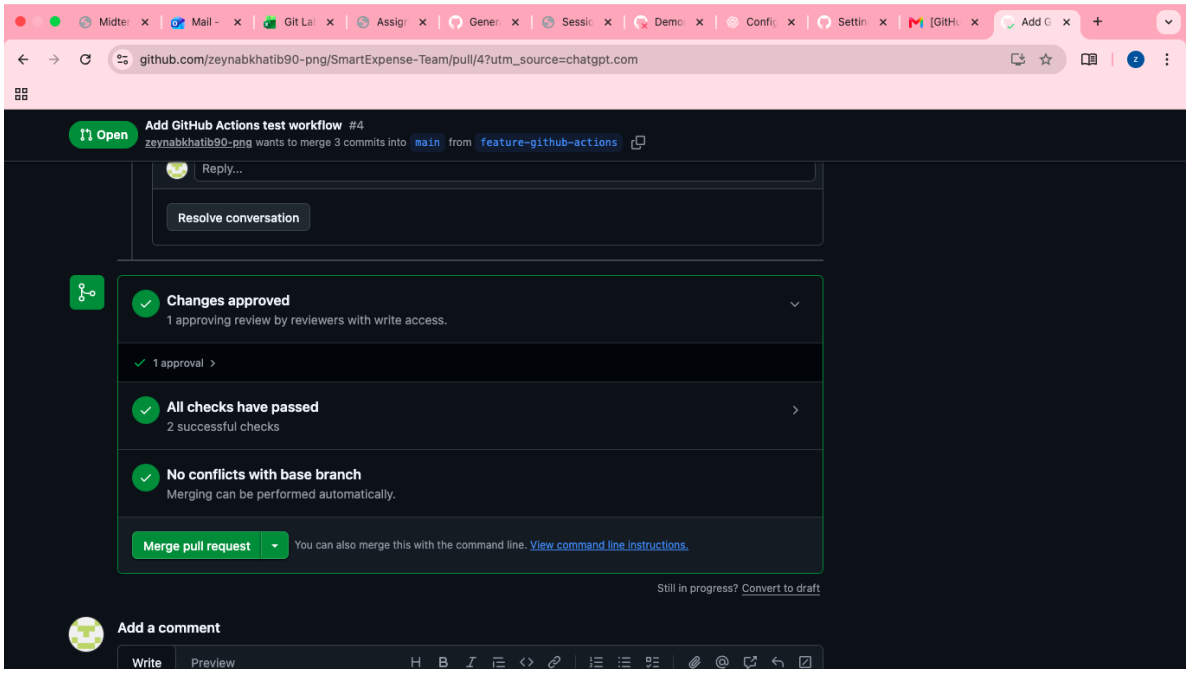
Zeynab: feature-expenses (expense management and tests, PR #1), feature-github-actions (CI workflow, PR #4), status-zeynab (README status, PR #8), and feature-pages-update (website update, PR #10). **Hiba:** feature-analysis (two commits for analysis and tests, PR #2), readme-status (PR #5), feature-pages (PR #7), and status-hiba (conflict-resolution PR #9). Both members worked using feature branches and Pull Requests.



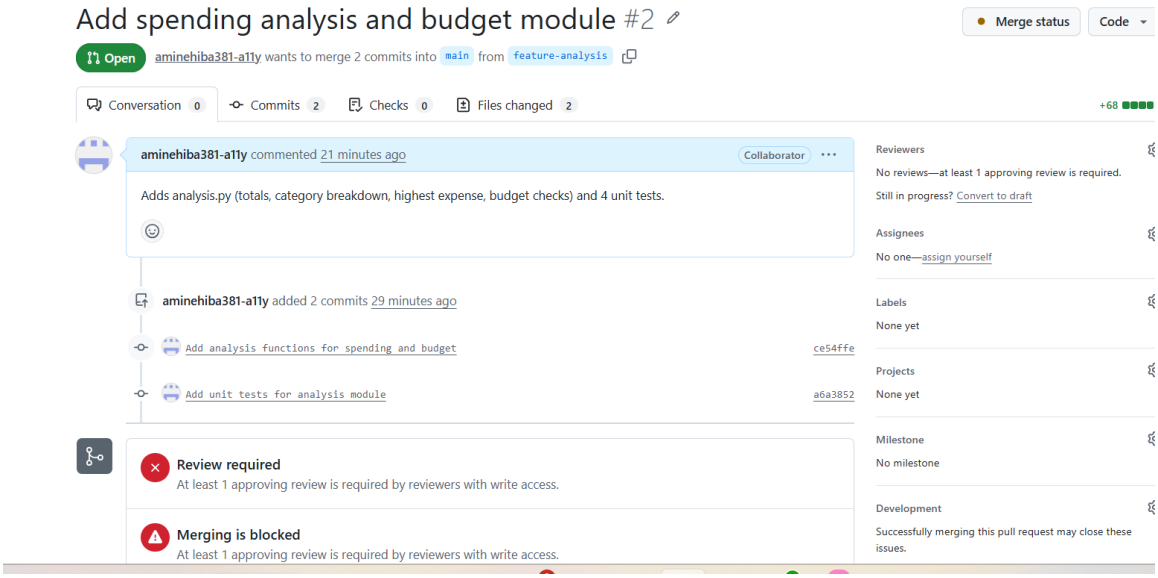
Evidence 2 - Hiba feature work and its GitHub review context; PR #2 is independently identifiable from the supplied evidence.

3. Pull Requests & Code Review

Both members reviewed the other member's changes. Hiba reviewed Zeynab's expense and workflow work; Zeynab reviewed Hiba's README, Pages and conflict-resolution changes. The screenshots establish meaningful line comments, approvals and required-review protection. A formal GitHub **Request changes** review followed by a correction on the same branch and reapproval is **not conclusively established** by these screenshots and should not be claimed as completed.



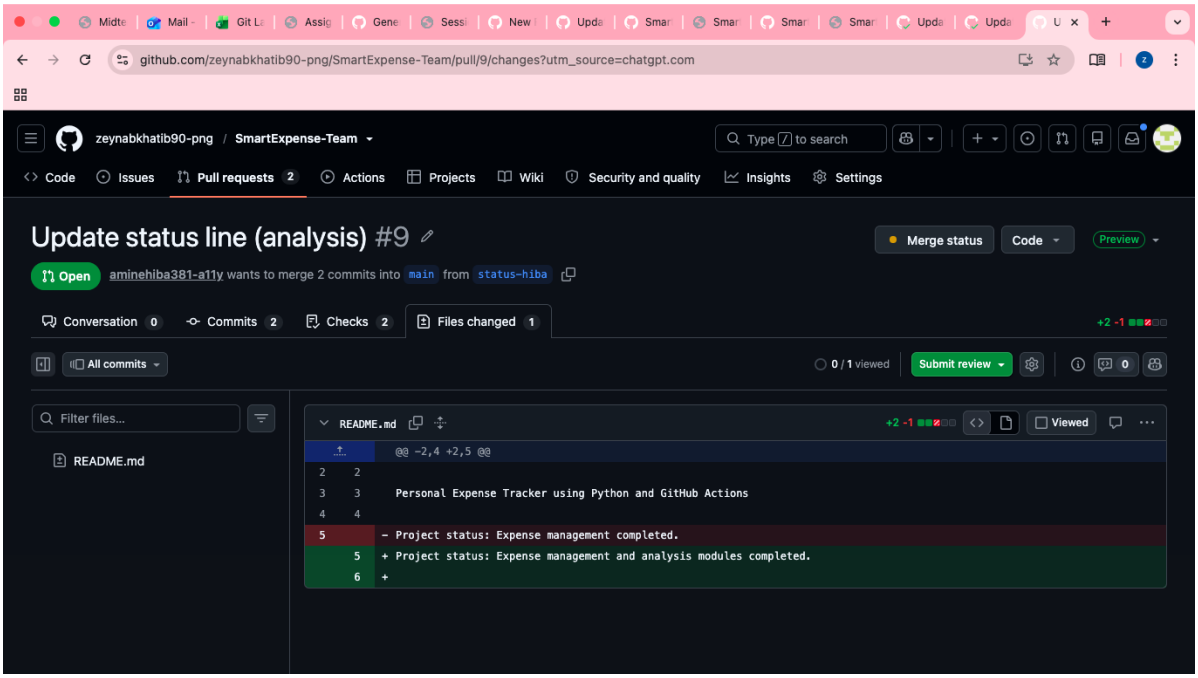
Evidence 3 - PR #4: approved review, successful checks and no conflicts before merging.



Evidence 4 - Hiba's code contribution and review context. Other line-level feedback is documented in the combined evidence.

4. Merge Conflict

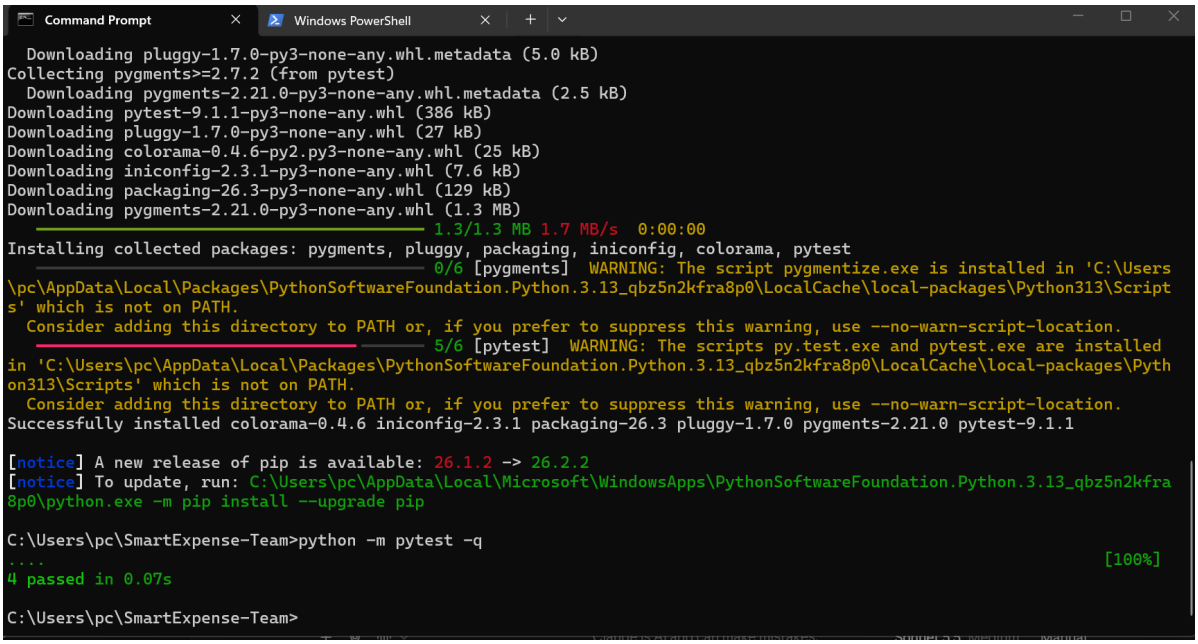
Both members started from the common README status line, **Project status: Development in progress.**, and edited it differently on separate branches. Zeynab's PR #8 merged first. Hiba then integrated updated main into status-hiba and committed **Resolve README merge conflict** (79bcd91). The final text was **Project status: Expense management and analysis modules completed.** PR #9 was reviewed and merged successfully.



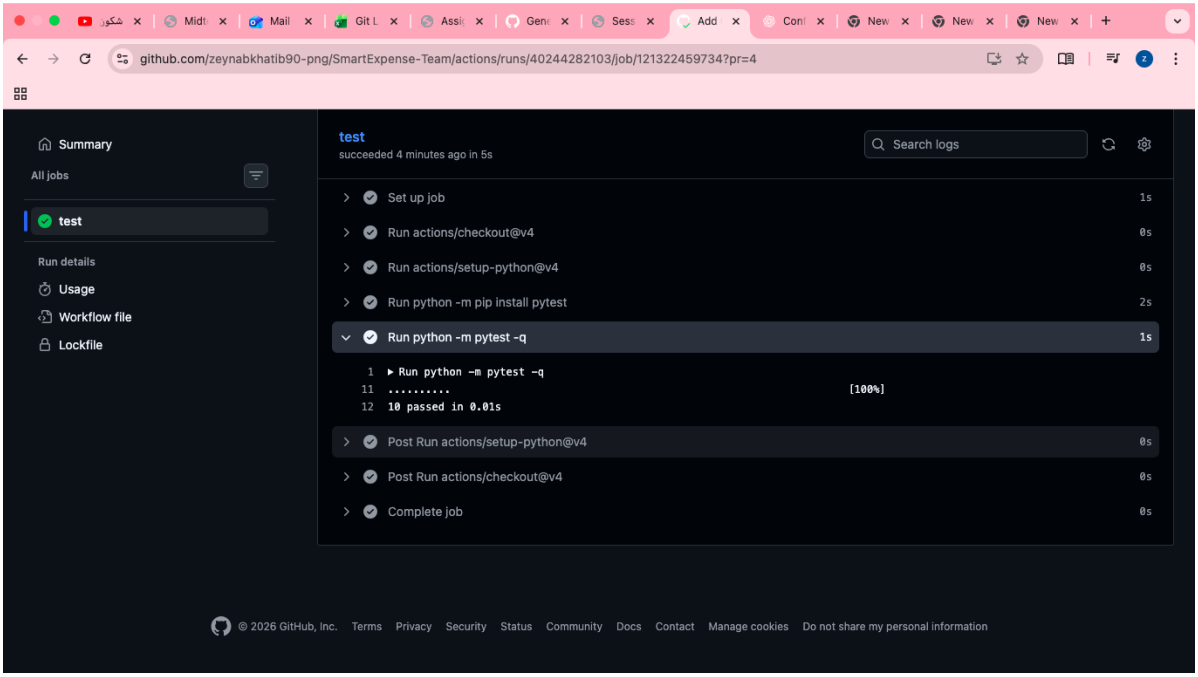
Evidence 5 - PR #9 diff shows the combined, resolved project-status line without conflict markers.

5. Unit Tests

The final GitHub Actions pytest output reports **10 passed**. Tests cover normal operations, boundaries and invalid inputs, including adding/removing expenses, invalid amounts and indexes, and analysis/budget behavior. Hiba separately supplied a local terminal screenshot documenting **4 passed** during an earlier development stage. This local result should not be presented as proof that the final ten-test suite ran locally.



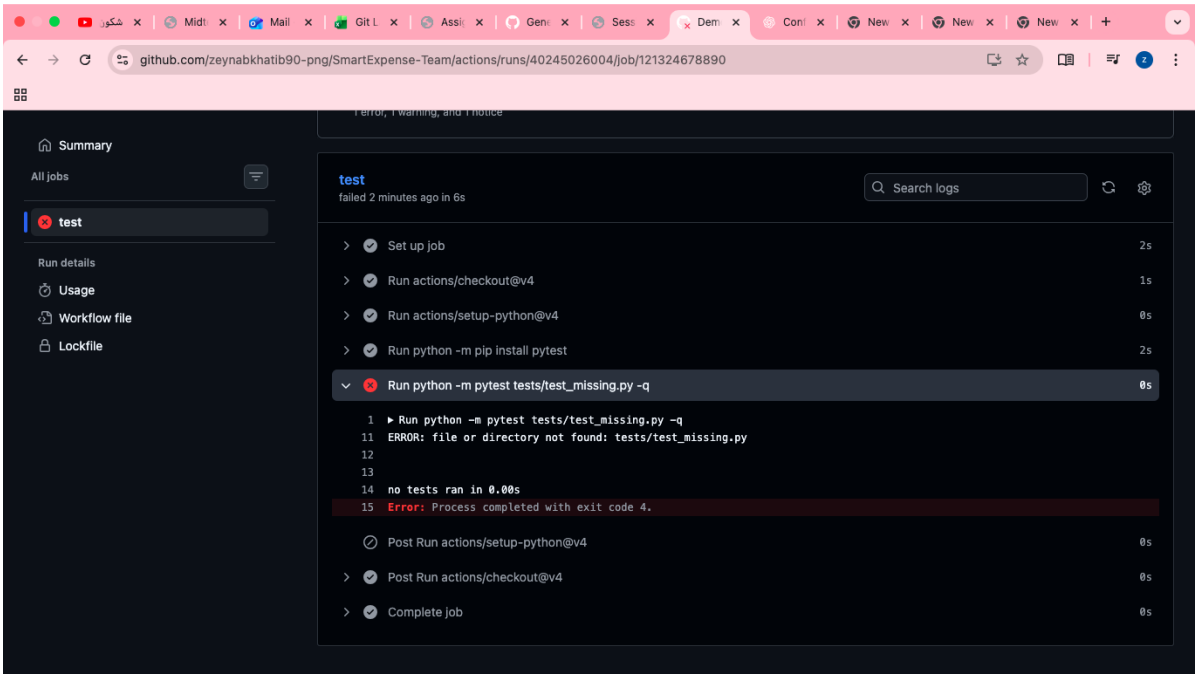
Evidence 6 - Hiba's local pytest run: 4 passed in 0.07s during development.



Evidence 7 - GitHub Actions running the full test suite: 10 passed.

6. Continuous Integration

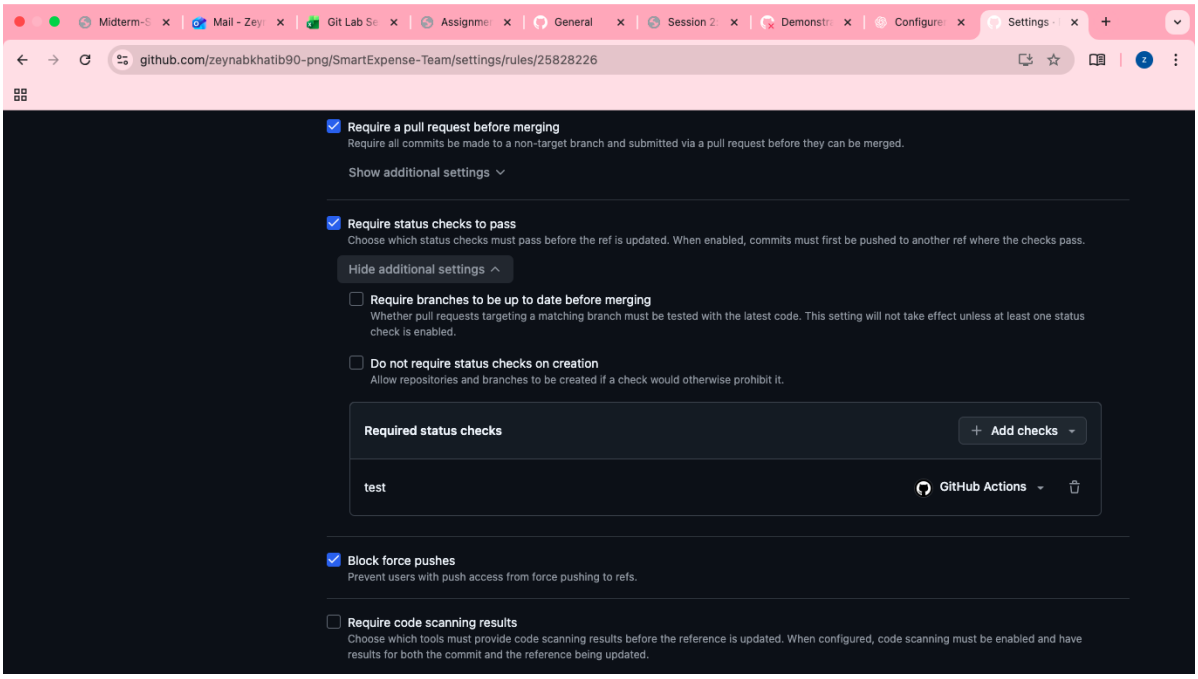
.github/workflows/tests.yml runs on pushes and pull requests. It uses an Ubuntu runner, Python 3.12, installs pytest, and runs **python -m pytest -q**. To demonstrate CI failure and recovery, a temporary test command referred to a nonexistent file, tests/test_missing.py. The job failed with exit code 4; the command was then fixed on the same feature branch and the complete test suite passed. This was an intentional CI demonstration, not a defect in expense calculations.



Evidence 8 - Failed CI log: nonexistent test file and process exit code 4.

7. Ruleset & Protected main

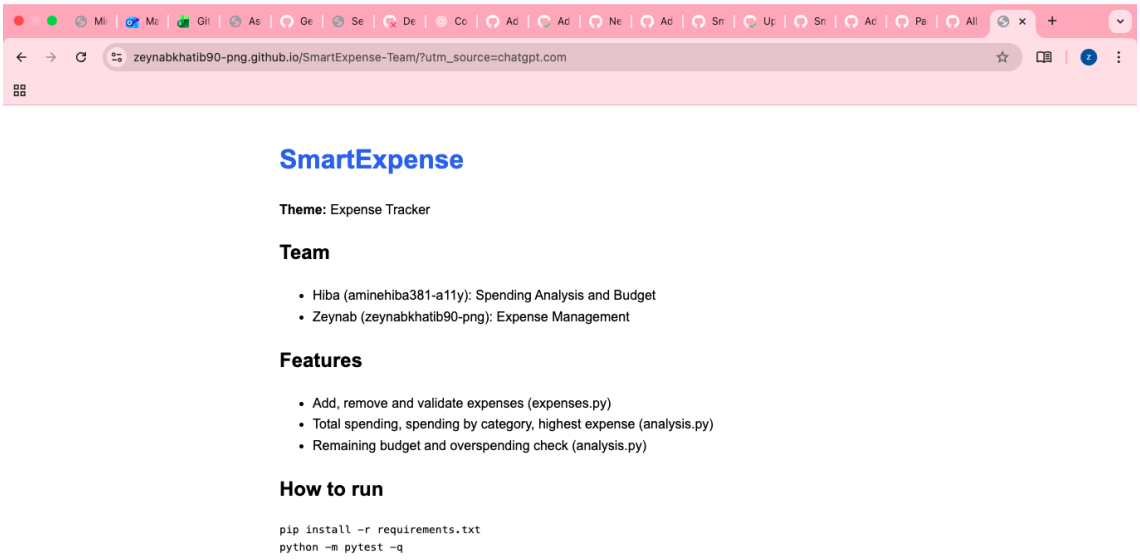
The active **Protect main** ruleset requires changes through pull requests, at least one approving review and a successful GitHub Actions **test** status check. Deletion and force pushes are restricted; the bypass list was empty in the supplied evidence. PR #4 demonstrated merge blocking until review and checks were satisfied.



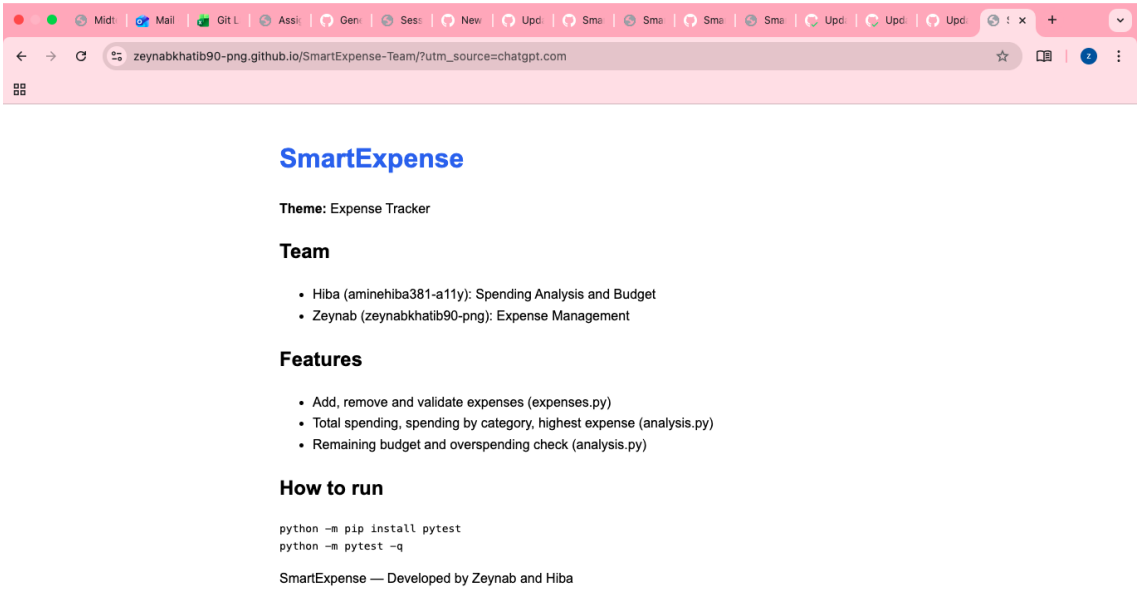
Evidence 9 - Required GitHub Actions test status check selected in main ruleset.

8. Continuous Deployment

The informational site at **docs/index.html** was deployed from **main /docs** using GitHub Pages. After the first deployment was cancelled and the site briefly returned a 404, a later deployment succeeded. PR #10 updated the site footer and installation instructions; the shared conversation includes a live screenshot showing the new footer **SmartExpense - Developed by Zeynab and Hiba** and the corrected pytest installation command. The before/after website states are documented below.



Evidence 10 - Original published SmartExpense project page, before the final footer update.



Evidence 11 - Published site after the PR #10 update, displaying the new footer and revised run command.

9. Contribution Summary

Zeynab Khatib	Expense management, five named pytest tests, PR #1/#4/#8/#10, CI failure/recovery, protected main and Pages deployment setup; reviewed Hiba's PRs.
Hiba Ait Salah	Spending analysis and budget logic, corresponding tests, PR #2/#5/#7/#9 and conflict-resolution commit; reviewed Zeynab's code/workflow.

10. Fair Use of AI

The group used ChatGPT to understand the GitHub interface, divide work, draft Python and pytest examples, configure GitHub Actions, troubleshoot rules and Pages, and help organize the report. Suggested outputs were adapted to the repository and checked through PR diffs, peer reviews, test logs and the published site. Both members remain responsible for understanding and verifying their own contributions. Each member should confirm that this accurately describes their use before submission.

11. Sources & Reflection

Sources: Assignment 2 - GitHub Collaboration & Automation (Group B); Session 2 slides; repository history and Pull Requests (#1, #2, #4, #5, #7, #8, #9, #10), GitHub Actions logs, protected-branch settings and the published GitHub Pages site.

The main challenges were managing protected branches, achieving a deliberate README conflict, and demonstrating CI failure then recovery. We learned to inspect changed files, reviewer state, automated checks and deployment logs before merging or diagnosing errors.

Final verification before submission: The uploaded evidence does not conclusively document the formal **Request changes - correction - reapproval** workflow or a local execution of the complete final ten-test suite. The assignment also explicitly requires **requirements.txt**, member/role details and the live Pages URL in README, and this final PDF committed to the public repository. Verify these repository items and submit the public repository URL in the shared sheet. Do not mark any unverified requirement as completed.